

From web game to **App Store and Play Store**, on our own VPS

The plain-words version of the launch plan for the team. Canonical source: LAUNCH_PLAN.md in the repo.

~52 dev days 12 to 14 weeks to store launch 16 decisions locked 14 phases

✅ **Locked.** Suhaib made every decision (D1 to D16) on 5 Sep 2026. The "Your decisions" tab now shows what was chosen and what it changes. Building starts in phase order.

The whole plan in 13 lines

- 1 Wrap, don't rewrite.** The React app ships inside a Capacitor shell. One codebase for web, iOS and Android.
- 2 Backend leaves Supabase and runs on our VPS, the palmandplate way.** Our own API (NestJS + Prisma) and a plain Postgres database on the Hostinger server we already run, behind the shared Traefik proxy, deployed by GitHub Actions over SSH. No Supabase software anywhere. Web and apps share that one database, so accounts, leaderboards, saves and "ads removed" are identical everywhere.
- 3 That same move fixes the Google login text.** The consent screen shows the callback domain. Ours becomes `api.letterlock.raltech.dev`, then we verify the brand so it reads "Letterlock".
- 4 Ads = Google AdMob.** Interstitial only between games, rewarded ads for an extra skip or hint, banner only on menus. Never during a question.
- 5 Remove Ads = one purchase, about \$3.99,** through Apple and Google billing via RevenueCat, mirrored to a flag on the account so it follows the login, not just the phone.
- 6 Purchases belong to the Apple ID or Google account by default.** Our server flag is what makes them follow the user across iPhone, Android and web. The "Remove Ads" tab explains it.
- 7 The song, TV-clip and iTunes-melody packs stay on the web and are hidden in the store builds.** Apple's guideline 5.2.5 forbids using iTunes previews "as the soundtrack to a game", and the number of clips does not change that. Self-made public-domain melodies grow to 200+ for both. Charades keep their images, from licensed, bundled sources.

- 8 Store requirements we do not have yet:** in-app account deletion, privacy policy and terms pages, Sign in with Apple (mandatory because we offer Google login), real PNG icons, offline handling.
- 9 "AI-looking code" is a myth.** Neither store checks who wrote the code. They reject thin web wrappers, crashes, placeholders and missing privacy features. The "Store review" tab is the real checklist.
- 10 Paperwork first:** Apple Developer (\$99/yr), Google Play (\$25 once), AdMob, RevenueCat, and a differentiated store name because exact "Letterlock" is already taken.
- 11 Ops we must add:** backups, uptime monitoring, crash reporting, a version gate, `app-ads.txt`, and over-the-air updates from day one so content waves never wait for a store review.
- 12 Realistic money:** ads earn tens of dollars a month at hundreds of daily players. The \$3.99 purchase will likely out-earn ads early. Do it for reach and polish, not quick revenue.
- 13 Mobile and TV are first-class targets.** A full mobile experience rehaul (portrait and landscape, no browser bar) built with the frontend-stack pipeline before submission, plus an Android TV / Google TV build controlled with the remote. See the "Mobile & TV" section.

Where we actually are today

AREA	REALITY	CONSEQUENCE
Web hosting	Cloudflare Pages, not the VPS. The VPS hosts nothing for Letterlock yet.	The migration builds the VPS side from scratch.
Backend	Supabase cloud project, ref committed in <code>.env.production</code> .	That ref is the random text on the Google consent screen. We leave it.
Backend footprint	Auth (email code via Resend, Google), 9 tables, ~30 database functions, 12 migrations, realtime broadcast + presence, 2 edge functions, no file storage.	Small and portable: about 30 endpoints and 3 realtime channels to rebuild in our own API.
Account deletion	None for users.	Store blocker on both stores.
Privacy policy, terms	None anywhere.	Store blocker, also needed by AdMob and Google brand verification.
Sign in with Apple	Not built.	Required by Apple 4.8 because Google login exists.
Icons	SVG favicon only.	Need 1024/512/192 PNG set and splash.
Join links	Query params only (<code>?room=</code>).	Add <code>/join/CODE</code> : universal links match paths reliably.
Remote media	1,624 iTunes song previews, 296 iTunes TV clips, ~600 CDN logos, random Flickr photos in charades.	Copyright, reliability and content-safety work (Copyright tab).
Monetization SDKs	Zero.	Ads, purchases, crash reporting are all new.

Decisions: locked on 5 Sep 2026

Every row is final. "Locked" is what Suhaib chose, "Means" is what the build does because of it.

#	DECISION	LOCKED	MEANS
D1	Store listing name	Letterlock: Party Quiz	Reserve it in both consoles the day the accounts exist. Bundle id stays <code>dev.raltech.letterlock</code> .
D2	Developer account	Organization: RAL Technologies on both stores (free D-U-N-S number first)	Company name on the listing; Play organization accounts skip the 12-testers / 14-day closed test.
D3	Song / TV / iTunes-melody packs	Web only. Hidden in the iOS, Android and TV builds.	Workaround checked: the clip <i>count</i> is irrelevant, the <i>source</i> (iTunes previews) is what Apple forbids, so no reduced version is store-safe. Packs are platform-gated; public-domain melodies grow to 200+ and ship everywhere.
D4	Ads placement and price	Interstitial after each game , rewarded for an extra skip or hint, banner on menus. Remove Ads \$3.99 one-time.	Never during a question, never on the board, never at app open.
D5	Static web hosting	Move to the VPS (PM2 behind Traefik). Cloudflare stays DNS only.	Cloudflare Pages retired at cutover. Everything Letterlock runs on our VPS.
D6	Target audience	13+, not directed at children	Kids pack stays, no child-directed marketing, standard ads allowed.
D7	Over-the-air updates	Yes, fully working at launch	Phase 3c: the Capgo updater plugin in manual mode pulling bundles from our own VPS (no Capgo account). Content reaches installed apps in minutes; native changes still go through the stores.
D8	VPS	The existing Hostinger KVM2 (2 vCPU, 8 GB RAM, 100 GB NVMe) that already runs palmandplate, jawhara and custompc	No new server. Details from those repos are in the "Off Supabase" section. Credentials only in gitignored files and GitHub secrets.
D9	Charades images	Keep images, make them clean and legal	Phase 1c: licensed sources (Pixabay, CC0, our own illustrations), human-reviewed, bundled in the apps, served from our domain on web, credits screen. No random Flickr photos.
D10	Logo packs	Keep : bundle SVGs, trademark disclaimer, drop the Apple logo, removal contact	How every Logo Quiz app operates.
D11	Fandom packs	Keep as text, "Unofficial fan trivia" label, rename the Harry Potter and Pokémon packs, never in store metadata	Facts are not copyrightable; the two most aggressive rights-holders get extra distance.
D12	Web ads	Yes, as the very last step , after the apps are live and AdMob is approved	Phase 8: AdSense "H5 Games Ads", same placements as the apps, honors the Remove Ads flag. Explained in the Ads tab.
D13	Web Remove-Ads purchase	No	A purchase made in an app still hides web ads for that account. No web checkout.
D14	What runs on the VPS	Our own backend, the palmandplate way. No Supabase software at all.	NestJS API + Prisma + the Postgres already on the box + Socket.IO realtime, PM2, Traefik, GitHub Actions over SSH. Phase 2 grows from 3 to about 14 dev days; users keep accounts, XP, friends and saves.

#	DECISION	LOCKED	MEANS
D15	TV scope at launch	Android TV / Google TV only	Same Android project, Play TV track, remote navigation, on-screen keyboard on inputs. LG and Samsung later as HTML5 apps.
D16	Mobile rehaul before submission	Yes, top priority: beautiful and fully playable	Phase 1b through the frontend-stack pipeline, before the store builds.

Also requested, now in the plan

ASK	WHERE
Web home screen pops up "Get the app" with store links, only after the apps are live	Phase 6b
Ads on the web too, at the very end	Phase 8, Ads tab
Over-the-air updates fully functioning at launch	Phase 3c
Full mobile rehaul, portrait and landscape, with frontend-stack	Phase 1b, Mobile & TV tab
TVs with the Play Store, remote control, keyboard pops on text fields	Phase 3b, Mobile & TV tab
VPS details and config from the jawhara / custompc / palmandplate repos, secrets gitignored	Off Supabase tab, Phase 2
Charades images kept and made to work properly	Phase 1c

Every phase, in build order

Phase 2 (the backend) is the critical path and runs in parallel with 1, 1b and 1c. Phase 0 starts today because it is mostly waiting.

0 Paperwork 1 to 2 weeks wait	1 Store-ready web 5 days	1b Mobile rehaul 6 days
1c Charades images 3 days	2 Our backend on the VPS 14 days + cutover	3 Capacitor apps 5 days
3b TV support 5 days	3c OTA updates 2 days	4 Ads 2 days
5 Remove Ads 3 days	6 Submission 2 days + 3 to 4 weeks	6b Get-the-app popup half a day
7 Ops 2 days, ongoing	8 Web ads 2 days, last	



Paperwork and accounts

you · 1 hour of forms · 1 to 2 weeks wait

- D-U-N-S number for RAL (free, 5 to 10 business days). Needed for Organization accounts.
- Apple Developer Program, \$99/yr, then Paid Apps Agreement, bank and W-8BEN tax form. Bahrain bank acceptance is unverified; a bank in another country is accepted if needed.
- Google Play Console, \$25 once, plus a Play merchant account. Bahrain is officially supported for both. Payout in USD by wire.
- AdMob account (Bahrain supported, \$100 payout threshold). Apps get approved only once live in a store.
- RevenueCat account (free under \$2,500/month).
- Google Cloud consent screen: app name Letterlock, logo, homepage, privacy URL, authorized domain, Search Console verification.
- Done on 5 Sep: name (D1) and VPS (D8) locked. Still needed from you: the D-U-N-S number, the store enrollments, and the VPS password in the gitignored `infra/vps-creds` file.



Store-readiness changes inside the web app

5 dev days

Everything here ships to the website too, so it pays off before the apps exist.

CHANGE	WHERE	WHY
Account deletion	New <code>delete_my_account()</code> database function, button in the profile view, web page for the Play listing	Apple 5.1.1(v), Play policy
Privacy policy + terms pages	Static pages on <code>letterlock.raltech.dev</code> , linked from Settings and both store listings	Both stores, AdMob, Google brand verification
Sign in with Apple	Apple provider in our auth server (Services ID, Team ID, .p8 key; secret renews every 6 months)	Apple 4.8
<code>/join/CODE</code> links	Router, QR code, copy link; <code>?room=</code> stays as alias	Universal links
Offline handling	Non-blocking banner, local play works offline, remote-media packs hidden offline, rooms disabled, score posts queued	Apple 2.1: reviewers test in airplane mode and IPv6-only
Version gate	<code>/api/app-config</code> with min version + maintenance flag	Retire old app versions, show maintenance notices
Icons + splash	1024 PNG master, generated sets, PWA icons	Stores + PWA
Bundle media	Flags, logo SVGs, fonts served locally	Offline, review reliability
Content changes	D3 (media packs web-only via a platform flag), D10, D11 applied; D9 is Phase 1c	Copyright tab
<code>app-ads.txt</code>	Root of the domain listed in the stores	AdMob requires it since Jan 2025



Mobile experience rehaul, portrait and landscape

6 dev days · frontend-stack

Playing on a phone is not great today, and in the browser the URL bar eats the screen in both orientations. This is a full optimization and rehaul of the mobile experience, run through the frontend-stack pipeline before any store screenshot is taken. Details in the "Mobile & TV" section.

- URL bar: the native app has none; the web version gets an "Add to Home Screen" prompt (full-screen PWA) and a Fullscreen button on Android Chrome; layout never shifts when the bar collapses.
- Landscape-first host layout: board left at max height, question and host pad right; slim header; thin timer bar. The phone controller locks to portrait.
- Touch targets at least 44 pt, primary actions in the thumb zone, destructive actions out of the way, type scale re-tuned per breakpoint.
- Keep the screen awake during a match; audio and motion tuned for low-end Android.
- Verified with the 17-device no-scroll matrix, before/after screenshots per orientation, and a real iPhone plus a mid-range Android.



Charades images, kept and made legal

3 dev days

Today the five charades packs (about 1,100 prompts) load a random Flickr photo at play time: unmoderated content in a family game, licenses we cannot honor, and a remote fetch that fails on reviewer networks. Suhaib wants the images kept, so the source changes.

- **Sources, in order:** Pixabay (safe search on, commercial use, no attribution needed), then Openverse filtered to CC0 / public domain, then our own generated flat illustrations for abstract prompts such as actions and emotions where photo search fails.
- **One script** fetches a candidate per prompt, resizes to 512 px WebP under 40 KB, writes a credits file per image, and only re-fetches what is missing.
- **Human review** on contact sheets before anything ships; a reject list forces a re-fetch or a generated fallback.
- **Bundled in the apps** (about 40 MB, fine for both stores) so the QR secret-prompt page works offline and on Apple's review network; served from our domain on web and cached.
- If an image is missing the word still shows, exactly as today. A build check refuses any pack with a remote image URL left in it.

Backend off Supabase, onto our VPS, the palmandplate way

14 dev days + a cutover evening

Will this continue for our app? Yes. The website, the phone apps and the TV build all talk to `api.letterlock.raltech.dev`, our own API. Same database, same accounts, same leaderboards, same saved games, same "ads removed" flag. After cutover there is no Supabase anything. The VPS facts, ports and paths are in the "Off Supabase" section.

What we copy from palmandplate and jawhara (same server, proven): NestJS + Prisma, the native Postgres, the email-code login service, refresh-token rotation, Resend for mail, PM2 processes, Traefik routing, GitHub Actions deploy over retrying SSH, a health check that gates every deploy, a Socket.IO realtime gateway, a nightly database dump. **What we add:** Google and Apple sign-in, guest tokens for phones that join a room without an account, an off-server backup copy, uptime monitoring.

STEP	DAYS	WHAT
Provision	0.5	Database role + two databases on the existing Postgres, folders, Traefik routing file, DNS records.
API skeleton	1	NestJS project, health endpoint, API docs page, environment-driven allowed origins, error filters.
Schema	1	Same 9 tables and the same ids as today, plus users, login codes, refresh tokens, guest tokens. The proven database functions (leaderboard paging, ranks, username cooldown, XP) stay as SQL and are called from the API.
Auth	3	Email code login (copied), Google on the web through our callback URL (this is the consent-screen fix), Google and Apple ID tokens from the apps, Sign in with Apple on web, account deletion, guest tokens, roles.
Data endpoints	3	About 30 routes replacing today's database calls: profile, username, leaderboard, ranks, XP, friends, saves, progress, custom packs, room members, admin. Every one of today's 23 row-security policies becomes an explicit "only your own rows" check , ticked off a checklist before cutover.
Realtime	2	Socket.IO gateway: token checked on connect, rooms assigned by the server. Mirrors today's lobby broadcast + presence, friends presence and notifications. Host reconnect re-sends the live question exactly as today.
Client rewrite	3	Remove the Supabase library. One API client with automatic token refresh, an auth provider, the lobby and friends code on Socket.IO. The existing username-gate and reconnect e2e suites must stay green.
Data migration	0.5	Dump the 9 tables plus user ids, emails and Google links; restore with the same ids so every link survives. Users sign in again once ; username, XP, friends, saves and leaderboard rows are all there.
CI/CD	1	Two workflows (main → prod, uat → dev): build, upload, SSH with 5 retries across two ports, apply migrations, restart PM2, poll the health check. Remove every Cloudflare Pages and Supabase step.
Backups + monitoring	(Phase 7)	Nightly dumps, 14-day local + 30-day off-server copy, Uptime Kuma. Done before cutover.
Cutover	evening	Final dump, restore, deploy, smoke test (code login, Google, Apple, room host + phone, leaderboard, resume), flip DNS. Supabase project kept two weeks as rollback, then deleted.

Why 14 days and not 3: running Supabase's open-source stack would have been a URL change. Suhaib chose no Supabase software at all, so login, data access and realtime are rewritten. The upside is one house pattern across RAL (palmandplate, jawhara, Letterlock), zero dependence on Supabase's roadmap, and a backend any RAL developer already knows.

3 The Capacitor apps

5 dev days

- Capacitor 8, bundle id `dev.raitech.letterlock`, web assets bundled inside the app (never loading the live URL).
- Plugins: app (deep links, Android back), browser, haptics, share, network, status bar, splash, keyboard, preferences, screen orientation.
- Native Google and Apple sign-in via ID tokens. Google blocks OAuth inside embedded web views, and the native sheet shows no domain text at all.
- Deep links: association files on our domain so QR codes open the app if installed, the website otherwise.
- Mirror saves and settings into native preferences (iOS can evict web storage). Handle inline video, first-tap audio unlock, no pinch zoom, Android 16 edge-to-edge.
- iOS builds on Codemagic (500 free macOS minutes a month) straight to TestFlight; Android bundles to the Play internal track. No Mac needed.

3b TV support: Android TV / Google TV with the remote

5 dev days

- Same Android project, TV-enabled: leanback launcher entry, touchscreen not required, 320×180 TV banner, 1080p TV screenshots, Android TV track review (D-pad navigable, focus always visible, no touch-only paths).
- Remote control: arrows move focus to the nearest control, Enter activates, Back closes the top modal. Arrow keys already move a cursor across the hex board.
- Text and numbers: focusing a field on Android TV opens the system on-screen keyboard; we keep TV inputs minimal (room code, username) and add a big remote-friendly number pad for room codes.
- 10-foot UI: big type, 5% overscan margin, high contrast, nothing reachable only by hover or scroll. No banner ads on TV.
- Works as the shared board with phones as controllers, or alone with the remote in host-adjudicated mode.
- Tested on the Android TV emulator and with a permanent keyboard-only Playwright run at 1920×1080.



Over-the-air updates, live at launch

2 dev days

- The Capgo updater plugin in **manual mode**, no Capgo account: the app asks our API for the latest bundle, downloads the zip from our VPS, checks the checksum, switches to it, and reports "ready". If the new bundle fails to boot within 10 seconds it rolls back by itself.
- CI zips the web build and publishes it on the same push that deploys the website, so web and installed apps run the same commit within minutes.
- OTA moves **JS, CSS, content and images only**. Anything native (a new plugin, a permission, an SDK bump) goes through the stores, and a minimum-native-version gate keeps old shells from loading a bundle they cannot run. Apple allows this (guideline 3.3.2).
- Two channels: production, and beta for RAL team phones so a wave is tested on real devices first.
- Fallback, decided now so nobody re-debates it: if the manual path is flaky in testing, switch to Capgo cloud (\$12/mo) with the same plugin.



Ads

2 dev days + AdMob approval 1 to 2 weeks after launch

- AdMob via the Capacitor community plugin. AdMob only, no mediation until real volume.
- Consent form on every launch before ads (EU/UK/US messages), a "Privacy options" entry in Settings, and the iOS tracking prompt with a short explainer. Declining is fine: ads stay, just untargeted.
- Placement: interstitial on the result screen and between games only, capped; rewarded ad for an extra skip or hint; banner on Home, Category menu, Settings, Lobby. Never on the board, never on the phone controller during a question.



Remove Ads purchase

3 dev days

- One non-consumable product in both stores, \$3.99 tier, Family Sharing on for iOS.
- RevenueCat SDK, entitlement `no_ads`, linked to our user id at sign-in, restore behavior "transfer".
- Restore Purchases button (Apple requires it). RevenueCat calls a webhook on **our API**, which writes `profiles.ads_removed` in our database; refunds revoke it automatically. The same flag hides web ads once Phase 8 ships.



Store submission

2 dev days + 2 to 4 weeks waiting

- Assets from the real build: iPhone 6.9" and iPad 13" screenshots, Play phone and tablet screenshots, 1024×500 feature graphic. No logos or franchise names in screenshots.
- Forms: App Privacy labels, the new age-rating questionnaire, Play Data safety, content rating, "Contains ads", target audience 13+.
- Reviewer notes with a demo room flow and test account. Budget one rejection round.



Web → app funnel, switched on after both stores approve

half a dev day

- **iPhone Safari:** Apple's own Smart App Banner (one meta tag), which also deep-links into the same room.
- **Every other phone browser:** a dismissible sheet on Home only: "Play Letterlock in the app" with the official App Store and Google Play badges. Mobile browsers only (never inside the apps, never on desktop or TV), remembered for 14 days after dismissal, never blocking the game.
- **Switch:** the store links come from the API and stay empty until both apps are live, so nothing shows before launch.
- **Rule:** the popup promotes the app and nothing else. No prices, no "cheaper here" (Apple 3.1.3).



Keep it alive

2 dev days, then ongoing

Backups first

Nightly dumps of both databases plus the env files, 14 days local and 30 days off-server, quarterly restore test. Neither palmandplate nor jawhara has the off-server copy yet; Letterlock ships it first.

Uptime

Uptime Kuma probing the site, the API health check, a realtime connect and the app-config endpoint, alerting your phone.

Crashes and errors

Sentry free tier plus the stores' own vitals. Play hides apps above a 1.09% crash rate.

Network in the app

Yes, we need it: a connectivity listener, an offline banner, queued writes, and a maintenance flag during VPS work.

Updates

Node, PM2 and Postgres upgrades are manual and announced; dependency audit in CI; fail2ban and a move to SSH keys. Yearly Play API and Xcode bumps. Apple sign-in secret every 6 months.

Takedowns

A published legal contact and a 48-hour removal routine for any logo or fandom complaint.



Web ads, the very last step

2 dev days + AdSense approval

- Only after the apps are live, AdMob is approved, and `app-ads.txt` / `ads.txt` sit on the domain.
- Apply for Google AdSense on the domain, then for the **H5 Games Ads** program (a second approval, made for browser games).
- Integration: the Ad Placement API, with a full-screen ad at the same moments as the apps (after a game) and a rewarded ad for the extra skip or hint. Google's built-in consent message covers the EU and UK.
- Same rules and the same Remove Ads flag: a signed-in buyer sees no web ads either. Never on the board, never during a question, never on TV.
- Reality: web game ad rates are about half of app interstitials. This is done for completeness because it costs two days, not for revenue.

Mobile and TV: first-class targets

Two additions to the plan: a full rehaul of the phone experience in both orientations, and a TV build you control with the remote.

Mobile: the problems and the fixes

PROBLEM	FIX
Browser URL bar steals space, portrait and landscape	Three layers, all shipped. 1. The native app has no browser chrome at all (the real fix). 2. PWA standalone: an in-app "Add to Home Screen" prompt (iOS Safari steps, Android install prompt) so the web version opens full-screen with no bar, today. 3. In the browser: a Fullscreen button on the board screen for Android Chrome (iPhone Safari does not allow it, so iPhone gets the PWA route), and a layout that never shifts when the bar collapses or expands.
Landscape is cramped	Landscape-first host layout: board left at max height, question and host pad right as one column, header collapses to a slim strip, timer becomes a thin top bar. The phone controller locks to portrait.
Touch targets and thumb reach	Every tappable at least 44 pt; host pad and the controller answer button anchored to the bottom thumb zone; exit and undo moved away from the primary thumb path.
Small text at arm's length	Type scale re-tuned per breakpoint; question text at least 18 px on phones; hex letters scale with the board.
Screen dims or locks mid-match	Keep-awake during a match (native plugin in the app, Wake Lock in Chrome).
Low-end Android jank	Motion budget: claim animation and winning trace stay, ambient particles reduced on data-saver and low-memory devices; measured with CPU throttling.
How we verify	The 17-device no-scroll matrix (including landscape phones) stays the gate, plus before/after screenshots per screen per orientation, plus a real iPhone and a mid-range Android.

Process: this phase runs through the frontend-stack pipeline: style pick, taste floor, guidelines review, Playwright verification. It happens before any store screenshot is taken (decision D16).

TV: which TVs have the Play Store

TV	PLAY STORE?	WHAT THAT MEANS FOR US
Philips (TP Vision models sold in Europe, Middle East, Asia)	Yes	Android TV / Google TV. Our Android build runs, downloaded from the Play Store on the TV.
Sony, TCL, Hisense (most), Sharp, Xiaomi, Nvidia Shield, Chromecast with Google TV, onn	Yes	Same.
LG	No	webOS with the LG Content Store. It accepts HTML5 apps, so the same web build can be packaged later.
Samsung	No	Tizen with Samsung TV Apps. Also accepts HTML5 apps; same later path.
Roku TVs	No	Not planned.
Apple TV	No	No web view, no Capacitor. AirPlay mirroring from an iPhone or iPad already shows the board.
Amazon Fire TV	Own store	Android-based; the Android TV build can go to the Amazon Appstore later.

TV: how it will work with the remote

AREA	WHAT WE DO
Store listing	Same Android project, TV-enabled: leanback launcher entry, touchscreen not required so one app serves phones, tablets and TVs, 320×180 TV banner, 1080p TV screenshots, Android TV track with its own review (D-pad navigable, focus always visible, no touch-only paths, no portrait, Back behaves).
Remote (D-pad)	Remote presses arrive as keyboard events. A spatial focus manager moves focus to the nearest control in the arrow's direction, Enter activates, Back closes the top-most modal or asks to exit. The hex board already supports arrow-key play. One big, high-contrast focus ring everywhere.
Typing text and numbers	Focusing a text field on Android TV pops the system on-screen keyboard automatically; Enter submits, Back dismisses. TV inputs stay minimal (room code, username), plus a big remote-friendly number pad for room codes. Nothing depends on hover.
10-foot UI	Body text at least 24 sp, headings at least 48 sp, 5% overscan margin, blue/amber palette (already TV-safe), no small chips, no scroll-only lists. No banner ads on TV; interstitials still only between games.
Roles on a TV	TV as the shared board while phones are controllers (best), or TV alone with the remote in host-adjudicated mode. Both must work with the remote only.
Testing	Android TV emulator for the real remote path, plus a permanent keyboard-only Playwright run at 1920×1080: every screen completable with arrows, Enter and Escape.

Off Supabase: what that means exactly

You said "we will migrate off Supabase" and then locked D14: our own backend, the palmandplate way. Here is precisely what leaves and what replaces it.

	TODAY	AFTER PHASE 2
Where the data lives	Supabase's servers (their cloud project)	Our VPS, plain Postgres
Supabase account, bill, software	Cloud project	None. No Supabase cloud, no Supabase open-source stack, no Supabase library in the app. Project deleted two weeks after cutover.
Domain users see	<code>lkudntyvngww1zuciocd.supabase.co</code>	<code>api.letterlock.raltech.dev</code>
Who can switch it off	Supabase	Only us
Server software	Supabase's hosted stack	Our NestJS API + Prisma + Postgres + Socket.IO, the same shape as palmandplate and jawhara
Login	Supabase Auth	Our own: email code, Google (web redirect and native token), Apple, 15-minute access tokens with rotating 90-day refresh tokens
Data access	Supabase client + row-level security	REST endpoints; every policy becomes an explicit "only your own rows" check in the API
Realtime	Supabase Realtime	Socket.IO gateway, token checked on connect, rooms assigned server-side
Email	Edge function → Resend	Resend over SMTP from the API, as palmandplate does

Our VPS, as documented in the palmandplate, jawhara and custompc repos

Read on 5 Sep 2026 from those three repos. Nothing here is a secret; passwords live only in gitignored files and GitHub Actions secrets.

FACT	VALUE
Provider, plan	Hostinger KVM2 : 2 vCPU, 8 GB RAM, 100 GB NVMe, about \$8.49 a month, shared by every RAL project
Host	<code>srv1167964</code> , IPv4 <code>72.62.16.1</code> , IPv6 available (the office network cannot reach the IPv4, so the helper scripts fall back to IPv6)
OS, access	Ubuntu 24, SSH on ports 22 and 2222, root with a password (moving to keys is a shared RAL TODO)
Free today	About 5 GB RAM and 63 GB disk
Reverse proxy	Traefik in Docker owning ports 80/443, Let's Encrypt certificates, hot-reloaded routing files in <code>/root/traefik-dynamic/</code>
Postgres	Native install, one instance on localhost, one role and database per project (palmandplate style). Jawhara runs its own Postgres containers on other ports.
Processes	palmandplate: PM2 (API + static sites). jawhara and custompc: Docker containers behind Traefik.
Deploy	GitHub Actions → SSH → apply migrations → build → restart. Jawhara retries SSH five times across both ports because Hostinger sometimes drops fresh connections from GitHub.
DNS	Cloudflare zone <code>raltech.dev</code> , plain A records to the VPS. DNS-only (grey cloud) until the first certificate exists.
Backups	jawhara: nightly dump, 14-day rotation, no off-server copy yet. palmandplate: none.
Other tenants (never touch)	palmandplate, jawhara, custompc, n8n, docuseal, onecli and its Postgres

What Letterlock adds to that server

ITEM	VALUE
Folders	<code>/opt/letterlock</code> (prod) and <code>/opt/letterlock-dev</code> (dev, from the <code>uat</code> branch)
Database	Two databases on the existing Postgres, <code>letterlock</code> and <code>letterlock_dev</code> , one role, a generated password pasted into GitHub secrets. Never reachable from outside the server.
Processes (PM2)	API on ports 3100 / 3101 bound to localhost only, static web on 5190 / 5191. Ports chosen to avoid everything already in use.
Domains	<code>letterlock.raltech.dev</code> , <code>api.letterlock.raltech.dev</code> , and <code>dev.</code> / <code>api.dev.</code> twins
Firewall	SSH, 80, 443 only. The API ports are never opened (palmandplate has its API port open to the world; we do not copy that).
Committed config (no secrets)	Traefik routing file, PM2 ecosystem file, server setup script, <code>.env.example</code> with variable names, both deploy workflows, the SSH helper script, the backup script
Gitignored secrets	<code>infra/vps-creds</code> , <code>infra/db-creds</code> , known-hosts pins, every <code>.env</code> . Unlike jawhara, Letterlock never commits credentials, private repo or not.
RAM	About 300 MB for the API plus the static servers. Postgres is already running.

Fixing the "xxxx.supabase.co will access" text

Why it happens: Google shows the domain of the login callback until the app's branding is verified. Today that callback is on the Supabase domain, so users see the random project ref.

- 1** Move the callback to `api.letterlock.raltech.dev`, served by our own API (Phase 2). The old supabase.co redirect is removed at cutover.
- 2** Verify `raltech.dev` in Google Search Console.
- 3** Google Cloud branding: app name Letterlock, logo, homepage, privacy and terms URLs on that same domain.
- 4** Set the audience to External and publish. Testing mode hides branding.
- 5** Click Verify Branding. Minutes to three business days. Result: "Sign in to continue to Letterlock" with our logo. In the apps the native Google sheet shows no domain at all.

How do we know who bought Remove Ads?

Short answer: the store account (Apple ID on iPhone, Google account on Android) owns the purchase. Our login does not, unless we copy it there ourselves. We do both.

- 1 Someone taps Remove Ads on an iPhone. Apple charges their Apple ID and records "this Apple ID owns remove_ads". Our account is invisible to Apple and Google.
- 2 Same Apple ID, new phone or reinstall: Restore Purchases re-grants it with no login of ours. Apple requires that button. Android re-grants automatically.
- 3 An iPhone purchase does not appear on Android, and neither appears on the website. Only our server can bridge them.
- 4 Receipts can be faked on a jailbroken phone, so the flag is written only after server-side verification. RevenueCat verifies with Apple or Google and calls our webhook, which sets `ads_removed = true` on the account in our database.
- 5 iPhone, Android and web all read that one flag when signed in. **Ads are hidden if either the store says this device owns it, or the signed-in account has the flag.**

The rule set

SITUATION	WHAT HAPPENS
Buys as a guest, signs in later	The anonymous purchase merges into the account at sign-in; flag set.
Second device, different Apple ID or Android	Store restore fails, but the account flag is true, so no ads. This is the whole point of the flag.
Two of our accounts share one Apple ID	The entitlement transfers to whoever restores last. One account at a time, so one purchase cannot unlock unlimited accounts.
Refund or chargeback	Apple or Google notify RevenueCat, the webhook sets the flag back to false.
Family Sharing (iOS only)	Family devices get it through the store; we do not write the flag to their accounts.
User deletes their account	Flag is gone. Restore Purchases re-creates it on that store account.
Wants to buy on the web	No web purchase (D13). A purchase made in an app hides web ads for that account through the same flag.

What we chose

CHOICE	PICKED	REJECTED
Billing	Apple In-App Purchase + Google Play Billing (mandatory for digital goods)	Stripe inside the app (instant rejection)
SDK	RevenueCat: free under \$2.5k/mo, does receipt checks, refunds, restore, cross-platform, webhooks	Free plugins where we host receipt verification, Apple notifications, Google Pub/Sub and reconciliation ourselves
Type	One-time, about \$3.99	Subscription (wrong vibe for a family game)
Web	Web ads as the last phase; the account flag hides them for buyers; no web purchase (D13)	Advertising web prices inside the iOS app (Apple 3.1.3 outside the US)

Ads: what to know

- **Placement rules are law, not taste.** Play forbids interstitials at app open, before a level, mid-action, or on exit. End of a game is the allowed natural break. Rewarded ads are always fine because the user opts in.
- **Consent is mandatory** in the EU and UK through Google's own consent SDK, and recommended everywhere.
- **AdMob approval happens after launch.** They check `app-ads.txt` on our domain and the live listing.
- **Money reality.** Revenue per 1,000 ad views: interstitial about \$10 in the US, \$5 in the EU, \$1 to \$4 in MENA; rewarded \$16 to \$20 in the US; banners about \$1. A party game shows two or three interstitials per half-hour session for a whole room on one device.

\$10 to \$40 PER MONTH AT 300 DAILY PLAYERS	\$100 to \$400 PER MONTH AT 3,000 DAILY PLAYERS	\$3.99 REMOVE ADS, LIKELY THE BIGGER LINE EARLY
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Where ads go

FORMAT	WHERE	NEVER
Interstitial	Result screen, between games in a best-of series, max 1 per 3 to 5 minutes	Mid-question, during the timer, at app open, on the host pad, on the phone controller while a question is live
Rewarded (opt-in)	Extra skip, a hint, unlock a category for one match	Anything that blocks the turn
Banner	Home, Category menu, Settings, Lobby waiting	The board screen (shared TV screen)
Web (Phase 8, last)	AdSense H5 Games Ads: full-screen after a game, rewarded for a skip or hint, same caps	Before the apps are live; on the board; during a question

Web ads explained (D12), and why they come last

- **Apps and websites use different Google products.** Apps use AdMob (a kit inside the app). Websites use AdSense (a script on the page). Same Google account, two separate approvals, two publisher ids, two payout lines.
- **Normal AdSense hates game pages.** Its reviewers look for text; a page that is one board and a few buttons gets "low value content" rejections. Google built a special program for this, **H5 Games Ads**, which shows full-screen ads between rounds and rewarded ads on request, like the app placements. AdSense approval first, then H5 program access.
- **Why last:** approval needs the finished privacy and terms pages and a polished live site; a rejected application slows later approvals, so we apply once when everything is ready; AdMob approval and the store listings must exist anyway for `app-ads.txt`, and both files live on the same domain; and money: web game rates are about half of app interstitials, and most traffic will move to the apps.
- **Consent:** Google's own consent message inside AdSense covers the EU and UK. Nothing else to build.
- **Remove Ads on web:** honored through the account flag, bought once in an app, gone everywhere. No web checkout (D13).

Copyright verdicts

Ads and a purchase make the app commercial, which changes what we can use.

CONTENT	LEGAL	STORE	VERDICT
Guess the Song (1,624 iTunes previews)	High	High	Web only (D3). Hidden in every store build. Apple 5.2.5: previews "may not be used for their entertainment value (e.g. the soundtrack to a game)"; iTunes API terms: promo use only, must show the track name and a buy link, which would spoil the answer. The clip count does not change this.
TV Show and Sitcom clips (296)	High	High	Web only (D3). Same terms.
Melody, iTunes part (~212)	High	High	Web only (D3).
Melody, our 23 synthesized public-domain tunes	Low	Low	Keep, bundle, grow to 200+: classical, folk, anthems, nursery, pre-1930 songs.
Logos (~600)	Medium	Medium	Keep with disclaimer, bundle the SVGs, drop the Apple logo, no logos in marketing, removal contact.
Flags	Low	Low	Bundle everything; Sports pack still hotlinks a flag CDN, switch to local.
Charades images (random Flickr today)	High	Medium	Keep the images, change the source (D9, Phase 1c): licensed (Pixabay, CC0, our own illustrations), human-reviewed, bundled in the apps, credits screen. Flickr removed entirely.
Fandom packs (17, text only)	Medium	Low	Keep with "Unofficial fan trivia" label, rename Harry Potter and Pokémon packs, franchise names never in store metadata, 48-hour takedown routine.
Movies, TV, Music text trivia	Low	Low	Keep.

Why there is no cheaper music path. Spotify removed preview URLs for new apps in 2024. Deezer's API is non-commercial only and stopped issuing tokens. Apple's MusicKit needs every player to hold an Apple Music subscription and forbids indirect monetization. Every surviving music-quiz app either licenses clips (SongPop) or gates behind Apple Music. Licensed clip vendors exist if music packs ever justify the cost.

"Will Apple ban it for looking AI-made?"

No. Neither store has any rule about who or what wrote the code, and reviewers never see the source. They judge the product. Here is what actually gets apps rejected, and our answer to each.

REAL REJECTION REASON	OUR ANSWER
4.2 "Repackaged website"	Bundle the build inside the app, native haptics, share sheet, deep links, native login, offline mode, in-app purchase.
4.3 Spam or clone (tightened June 2026)	Original mechanic, 40k questions, differentiated name, original icon, honest screenshots.
2.1 Crashes, blank screens, placeholders, IPv6-only failure	Offline banner, hostnames only, no test ads, every URL live.
2.3 Metadata mismatch	Screenshots from the real build; dev seams already inert in production.
5.1.1 No privacy policy or account deletion	Phase 1 adds both.
4.8 Google login without an equivalent private option	Sign in with Apple.
3.1.1 Digital goods outside in-app purchase	Native purchases only, no web pricing mentioned in-app.
5.2 Intellectual property	Copyright tab changes.
Play: personal account skipped the closed test	Organization account, or start the 14-day test early.

Native-feel checklist before submitting

- Splash and icon, styled status bar, no browser chrome, no pinch zoom, no text selection on game surfaces, no link previews.
- Android back button always does something sensible: close modal, exit confirm, go home.
- Safe areas on notch phones and Android 16 edge-to-edge; landscape phones verified with the existing no-scroll checker.
- Haptics through the native plugin, native share sheet, in-app purchase, offline path.
- Privacy policy, support page, in-app account deletion, Sign in with Apple.
- Honest metadata, screenshots from the real build, age rating, privacy labels, data safety form, 13+ audience.
- Reviewer notes: demo room code, test account, how to play host and player.

Costs, timeline, risks

Costs

ITEM	COST
Apple Developer Program	\$99 per year
Google Play Console	\$25 once
D-U-N-S number	Free
AdMob, AdSense, RevenueCat under \$2.5k/mo, Sentry, Uptime Kuma, Codemagic 500 min/mo, our own backend, OTA from our VPS	Free
VPS	Already paid: the shared Hostinger KVM2, about \$8.49/mo across all RAL projects. Letterlock adds about 300 MB of RAM.
Offsite backups	\$1 to \$2 per month
Store cut	15% of purchases
Optional later	Capgo cloud \$12/mo only if manual OTA misbehaves, licensed music clips (quote). Pixabay API is free.

Timeline

PHASE	DEV DAYS	WALL-CLOCK	DEPENDS ON
0 Paperwork	0.5 + your forms	1 to 2 weeks	You
1 Store-ready web app	5	Weeks 1 to 2	Nothing, can start now
1b Mobile experience rehaul	6	Weeks 2 to 3	Phase 1
1c Charades images	3	Week 3	Nothing
2 Our backend on the VPS	14 + cutover	Weeks 1 to 6, in parallel	VPS access
3 Capacitor apps	5	Weeks 6 to 7	Phases 1, 1b, 2, Apple account
3b TV support (Android TV, remote)	5	Weeks 7 to 8	Phase 3
3c OTA updates	2	Week 8	Phases 2 and 3
4 Ads (AdMob)	2	Weeks 8 to 9	Phase 3, AdMob account
5 Remove Ads	3	Week 9	Phase 3, store products
6 Submission	2 + waiting	Weeks 10 to 13	Everything above, TV track review
6b Get-the-app popup	0.5	After both approvals	Phase 6
7 Ops	2	Week 2 onward	Phase 2
8 Web ads (AdSense H5)	2 + approval	After launch, last	Phases 4 and 6
Total	~52	12 to 14 weeks to store launch, web ads after	

Risks and unverified items

- Apple accepting a Bahraini bank for payouts is unverified. Fallback: a bank in another country or the company's bank.
- The custom backend is the critical path (14 days, touches login, data and realtime). It starts first and runs in parallel with the UI phases; the two-client reconnect tests must pass against the new API before cutover; the Supabase project stays two weeks as rollback.
- One shared server for every RAL project. Letterlock binds its API to localhost, uses its own database role, its own process names and ports, and never touches another tenant's files or routing.
- AdSense may reject the site. We apply once, late, with finished pages; web ads are a nice-to-have.
- Exact "Letterlock" name is taken. Reserve the chosen name as soon as the Apple account exists.
- Apple universal links with query strings are unreliable, hence `/join/CODE`.
- Anti-steering rules in the US and EU keep changing. We never mention web purchases in-app, so changes cannot hurt us.
- MENA ad rates are low. Remove Ads is the revenue line.
- No managed backups on the VPS. Backups are the first ops task, before cutover.